Schedule

- The following will be announced this week:
  - Case exam grades
  - Statistics on the updated midterm scores:
    - Scores of those having a question on F-test=current score*100/95
- Lab 6 material is on LearnHub and will be covered this Thursday

39

Agenda

- Hypothesis tests for:
  - Variances of two populations
- Correlation and simple regression

40

Corollary

If $\sigma_1^2 = \sigma_2^2$, $\quad \frac{S_1^2}{S_2^2} \sim F_{m-1,n-1}$.

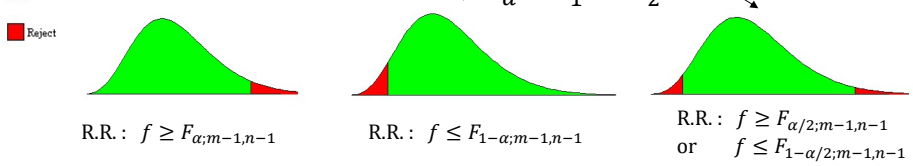
41

41

Tests about the equality of two population variances (or standard deviations) of two Normal random variables

H_0: $\sigma_1^2 = \sigma_2^2$ $\qquad$ Test statistics: $F = \frac{S_1^2}{S_2^2}$

H_a: $\sigma_1^2 > \sigma_2^2$ $\qquad$ H_a: $\sigma_1^2 < \sigma_2^2$ $\qquad$ H_a: $\sigma_1^2 \neq \sigma_2^2$



R.R. : $f \geq F_{\alpha;m-1,n-1}$ $\qquad$ R.R. : $f \leq F_{1-\alpha;m-1,n-1}$ $\qquad$ R.R. : $f \geq F_{\alpha/2;m-1,n-1}$ or $f \leq F_{1-\alpha/2;m-1,n-1}$

42

Example

- 12 oz bottles by a new & an old machine

Data

new: $s_1 = 0.1092$ $m = 15$

old: $s_2 = 0.12$ $n = 16$

Hypotheses Test?

$H_0: \sigma_1^2 = \sigma_2^2$

$H_a: \sigma_1^2 < \sigma_2^2$

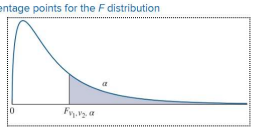
For $\alpha = 0.01, 0.10$ and α=0.05?

- Critical value? $F_{14,15,0.95}=?$

Critical value?

$F_{14,15,\,0.95}=?$

TABLE A.8 Upper percentage points for the *F* distribution



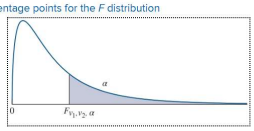
		ν_1								
ν_2	α	10	12	15	20	25	30	40	50	60
10	0.100	2.32	2.28	2.24	2.20	2.17	2.16	2.13	2.12	2.11
	0.050	2.98	2.91	2.85	2.77	2.73	2.70	2.66	2.64	2.62
	0.010	4.85	4.71	4.56	4.41	4.31	4.25	4.17	4.12	4.08
	0.001	8.75	8.45	8.13	7.80	7.60	7.47	7.30	7.19	7.12
11	0.100	2.25	2.21	2.17	2.12	2.10	2.08	2.05	2.04	2.03
	0.050	2.85	2.79	2.72	2.65	2.60	2.57	2.53	2.51	2.49
	0.010	4.54	4.40	4.25	4.10	4.01	3.94	3.86	3.81	3.78
	0.001	7.92	7.63	7.32	7.01	6.81	6.68	6.52	6.42	6.35
12	0.100	2.19	2.15	2.10	2.06	2.03	2.01	1.99	1.97	1.96
	0.050	2.75	2.69	2.62	2.54	2.50	2.47	2.43	2.40	2.38
	0.010	4.30	4.16	4.01	3.86	3.76	3.70	3.62	3.57	3.54
	0.001	7.29	7.00	6.71	6.40	6.22	6.09	5.93	5.83	5.76
13	0.100	2.14	2.10	2.05	2.01	1.98	1.96	1.93	1.92	1.90
	0.050	2.67	2.60	2.53	2.46	2.41	2.38	2.34	2.31	2.30
	0.010	4.10	3.96	3.82	3.66	3.57	3.51	3.43	3.38	3.34
	0.001	6.80	6.52	6.23	5.93	5.75	5.63	5.47	5.37	5.30
14	0.100	2.10	2.05	2.01	1.96	1.93	1.91	1.89	1.87	1.86
	0.050	2.60	2.53	2.46	2.39	2.34	2.31	2.27	2.24	2.22
	0.010	3.94	3.80	3.66	3.51	3.41	3.35	3.27	3.22	3.18
	0.001	6.40	6.13	5.85	5.56	5.38	5.25	5.10	5.00	4.94
15	0.100	2.06	2.02	1.97	1.92	1.89	1.87	1.85	1.83	1.82
	0.050	2.54	2.48	2.40	2.33	2.28	2.25	2.20	2.18	2.16
	0.010	3.80	3.67	3.52	3.37	3.28	3.21	3.13	3.08	3.05
	0.001	6.08	5.81	5.54	5.25	5.07	4.95	4.80	4.70	4.64
16	0.100	2.03	1.99	1.94	1.89	1.86	1.84	1.81	1.79	1.78
	0.050	2.49	2.42	2.35	2.28	2.23	2.19	2.15	2.12	2.11

Critical value?

$F_{14,15,\,0.95}=?$

$$= \frac{F_{14,15,0.95}}{1} = \frac{1}{F_{15,14,0.05}} = \frac{1}{2.46} = 0.407$$

TABLE A.8 Upper percentage points for the F distribution



		ν_1								
ν_2	α	10	12	15	20	25	30	40	50	60
10	0.100	2.32	2.28	2.24	2.20	2.17	2.16	2.13	2.12	2.11
	0.050	2.98	2.91	2.85	2.77	2.73	2.70	2.66	2.64	2.62
	0.010	4.85	4.71	4.56	4.41	4.31	4.25	4.17	4.12	4.08
	0.001	8.75	8.45	8.13	7.80	7.60	7.47	7.30	7.19	7.12
11	0.100	2.25	2.21	2.17	2.12	2.10	2.08	2.05	2.04	2.03
	0.050	2.85	2.79	2.72	2.65	2.60	2.57	2.53	2.51	2.49
	0.010	4.54	4.40	4.25	4.10	4.01	3.94	3.86	3.81	3.78
	0.001	7.92	7.63	7.32	7.01	6.81	6.68	6.52	6.42	6.35
12	0.100	2.19	2.15	2.10	2.06	2.03	2.01	1.99	1.97	1.96
	0.050	2.75	2.69	2.62	2.54	2.50	2.47	2.43	2.40	2.38
	0.010	4.30	4.16	4.01	3.86	3.76	3.70	3.62	3.57	3.54
	0.001	7.29	7.00	6.71	6.40	6.22	6.09	5.93	5.83	5.76
13	0.100	2.14	2.10	2.05	2.01	1.98	1.96	1.93	1.92	1.90
	0.050	2.67	2.60	2.53	2.46	2.41	2.38	2.34	2.31	2.30
	0.010	4.10	3.96	3.82	3.66	3.57	3.51	3.43	3.38	3.34
	0.001	6.80	6.52	6.23	5.93	5.75	5.63	5.47	5.37	5.30
14	0.100	2.10	2.05	2.01	1.96	1.93	1.91	1.89	1.87	1.86
	0.050	2.60	2.53	2.46	2.39	2.34	2.31	2.27	2.24	2.22
	0.010	3.94	3.80	3.66	3.51	3.41	3.35	3.27	3.22	3.18
	0.001	6.40	6.13	5.85	5.56	5.38	5.25	5.10	5.00	4.94
15	0.100	2.06	2.02	1.97	1.92	1.89	1.87	1.85	1.83	1.82
	0.050	2.54	2.48	2.40	2.33	2.28	2.25	2.20	2.18	2.16
	0.010	3.80	3.67	3.52	3.37	3.28	3.21	3.13	3.08	3.05
	0.001	6.08	5.81	5.54	5.25	5.07	4.95	4.80	4.70	4.64
16	0.100	2.03	1.99	1.94	1.89	1.86	1.84	1.81	1.79	1.78
	0.050	2.49	2.42	2.35	2.28	2.23	2.19	2.15	2.12	2.11

45

Example

- 12 oz bottles by a new & an old machine

Data

new: $s_1 = 0.1092$ $m = 15$

old: $s_2 = 0.12$ $n = 16$

Hypotheses Test?

$H_0: \sigma_1^2 = \sigma_2^2$

$H_a: \sigma_1^2 < \sigma_2^2$

For $\alpha = 0.01, 0.10$ and α=0.05?

- Critical value?
- Test statistics, F? F=0.8281
- p-value? P-value>0.10

$F_{14,15,\,0.99}=0.273$

$F_{14,15,\,0.95}=0.407$

$F_{14,15,\,0.90}=0.498$

46

Remarks

1. Normality is crucial. AT LEAST AN APPROXIMATE BELL SHAPE IS NEEDED FROM THE HISTOGRAM OF THE DATA.
2. Tests perform best when m=n. If m and n are very different and any doubt about normality DO NOT USE THE TESTS.
3. TESTS ARE NOT POWERFUL : H_0 will not be rejected fairly often even if H_0 is not true. Higher α can be a solution, α can be as high as 0.2 for these tests.
4. Type-II error, and sample size choice → Software packages...

47

47

Class Exercise

48

Class Exercise

	sample size	sample mean	Sample std. deviation
Line 1	12	5.2	0.5
Line 2	16	5.6	0.6

- The production manager wants to compare the standard deviations of the weights of popcorn per package from two production lines. Experience shows that the weights of popcorn can be assumed to be normal. He would like to know whether the true standard deviations of the weights of popcorn from the two production lines are different than each other. He obtained the above data:

$$H_0: \ \sigma_1 = \sigma_2$$
$$H_a: \ \sigma_1 \neq \sigma_2$$

- Formulate the corresponding hypotheses test. Do you think the true standard deviations of the weights of popcorn from two production lines are different at a confidence level of 90%? At 98% or 99.8%? Why or why not? What can you say about the p-value of the test?

49

Class Exercise

Hypotheses Test?

$$H_0: \sigma_1^2 = \sigma_2^2$$
$$H_a: \sigma_1^2 \neq \sigma_2^2$$

$s_1 = 0.5 \quad m = 12$
$s_2 = 0.6 \quad n = 16$

For confidence level 90%?

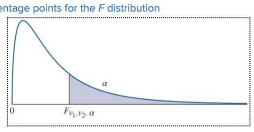
Critical values?

$F_{11,15,\,0.05} = 2{,}51$

$F_{11,15,\,0.95} = 1/\,2.72 = 0.368$

Test statistics, $F = \frac{0.5^2}{0.6^2} = 0.694$

TABLE A.8 Upper percentage points for the *F* distribution



Page 834

ν_2	α	ν_1 = 10	12	15	20	25	30	40	50	60
10	0.100	2.32	2.28	2.24	2.20	2.17	2.16	2.13	2.12	2.11
	0.050	2.98	2.91	2.85	2.77	2.73	2.70	2.66	2.64	2.62
	0.010	4.85	4.71	4.56	4.41	4.31	4.25	4.17	4.12	4.08
	0.001	9.75	9.45	9.12	8.80	8.60	8.47	8.28	8.19	8.12
11	0.100	2.25	2.21	2.17	2.12	2.10	2.08	2.05	2.04	2.03
	0.050	2.85	2.79	2.72	2.65	2.60	2.57	2.53	2.51	2.49
	0.010	4.54	4.40	4.25	4.10	4.01	3.94	3.86	3.81	3.78
	0.001	7.92	7.63	7.32	7.01	6.81	6.68	6.52	6.42	6.35
12	0.100	2.19	2.15	2.10	2.06	2.03	2.01	1.99	1.97	1.96
	0.050	2.75	2.69	2.62	2.54	2.50	2.47	2.43	2.40	2.38
	0.010	4.30	4.16	4.01	3.86	3.76	3.70	3.62	3.57	3.54
	0.001	7.29	7.00	6.71	6.40	6.22	6.09	5.93	5.83	5.76
13	0.100	2.14	2.10	2.05	2.01	1.98	1.96	1.93	1.92	1.90
	0.050	2.67	2.60	2.53	2.46	2.41	2.38	2.34	2.31	2.30
	0.010	4.10	3.96	3.82	3.66	3.57	3.51	3.43	3.38	3.34
	0.001	6.80	6.52	6.23	5.93	5.75	5.63	5.47	5.37	5.30
14	0.100	2.10	2.05	2.01	1.96	1.93	1.91	1.89	1.87	1.86
	0.050	2.60	2.53	2.46	2.39	2.34	2.31	2.27	2.24	2.22
	0.010	3.94	3.80	3.66	3.51	3.41	3.35	3.27	3.22	3.18
	0.001	6.40	6.13	5.85	5.56	5.38	5.25	5.10	5.00	4.94
15	0.100	2.06	2.02	1.97	1.92	1.89	1.87	1.85	1.83	1.82
	0.050	2.54	2.48	2.40	2.33	2.28	2.25	2.20	2.18	2.16
	0.010	3.80	3.67	3.52	3.37	3.28	3.21	3.13	3.08	3.05
	0.001	6.08	5.81	5.54	5.25	5.07	4.95	4.80	4.70	4.64
16	0.100	2.03	1.99	1.94	1.89	1.86	1.84	1.81	1.79	1.78
	0.050	2.49	2.42	2.35	2.28	2.23	2.19	2.15	2.12	2.11

50